Comparing and combining molecular activities inferred from different omics modalities

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The study:
TGF-beta perturbation on a kidney mesenchymal cell line

Article

SOURCE DATA · TRANSPARENT PROCESS · OPEN ACCESS · CHECK FOR UPDATES

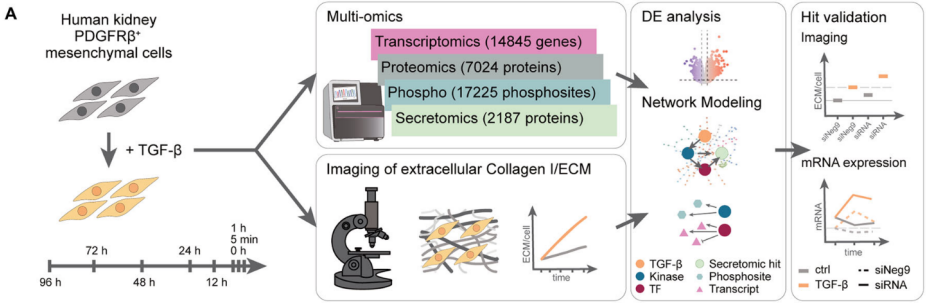
molecular systems biology

Dynamic multi-omics and mechanistic modeling approach uncovers novel mechanisms of kidney fibrosis progression

Nadine Tuechler[1,2,3,4,9], Mira Lea Burtscher[1,3,4,9], Martin Garrido-Rodriguez[1,3,5], Muzamil Majid Khan[1,6], Dénes Türei[3], Christian Tischer[1], Sarah Kaspar[1], Jennifer Jasmin Schwarz[1], Frank Stein[1], Mandy Rettel[1], Rafael Kramann[7,8], Mikhail M Savitski[1,2], Julio Saez-Rodriguez[2,3,5]✉ & Rainer Pepperkok[1,2]✉

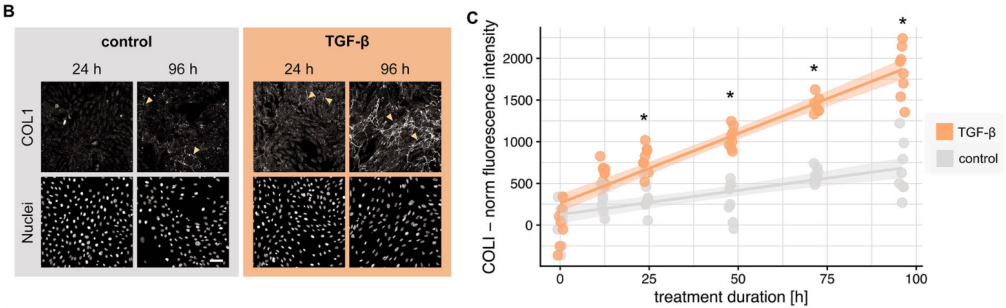
- Tuchler, Burtscher et al., Mol Sys Biol 21(8):1030-1065 (2025).
- Human kidney PDGFRβ+ mesenchymal cell line
- TGF-β stimulation: fibrosis model
- Collagen production: ECM deposition proxy

Overview of the study

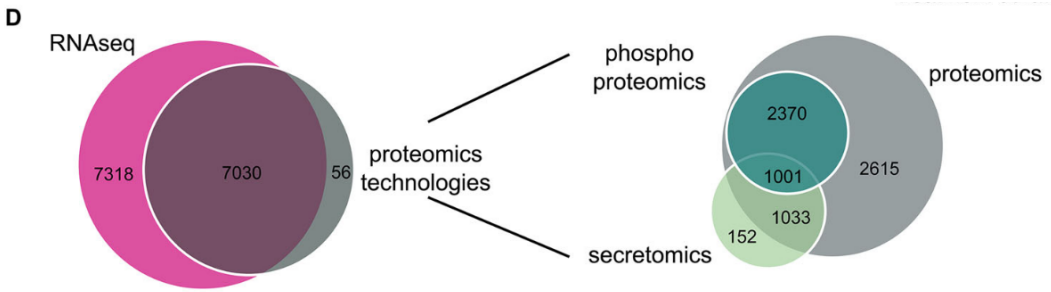


- Longitudinal setup: 7 time points
- Transcriptomics, proteomics, phosphoproteomics, secretomics
- Imaging of collagen
- DE analysis
- Mechanistic modeling

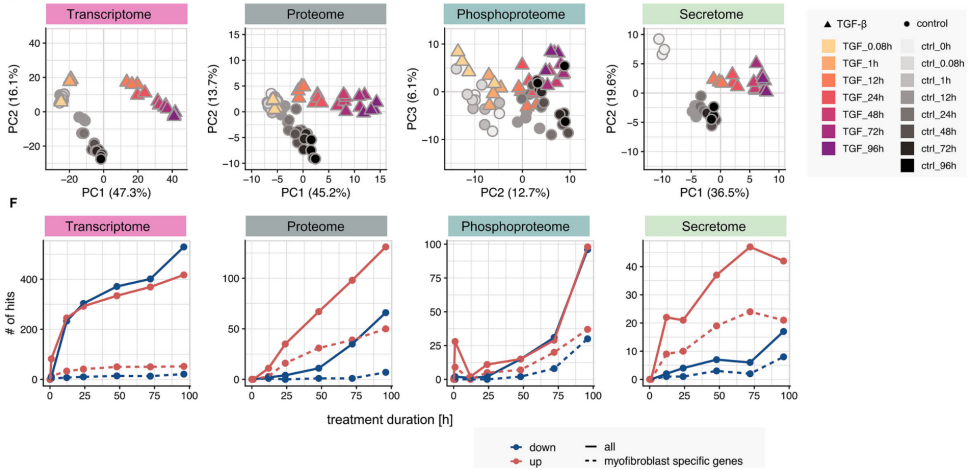
Collagen production upon TGF-beta stimulation



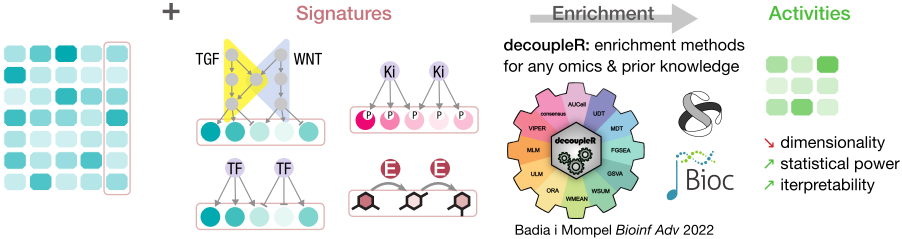
Transcripts and proteins measured



PCA of samples and number of DE transcripts or genes by omics modality



Activity inference using prior knowledge of omics signatures



Badia i Mompel *Bioinf Adv* 2022

Network modeling by causal reasoning

Activities

Pathway, TF, kinase, metabolic enzyme...

+

Network knowledge

- Causal, direct interactions
- 160k PPI

Turei *Nature Methods* 2016
Turei *Mol Sys Bio* 2021

Optimization

Causal reasoning: CORNETO, CARNIVAL
Boolean modeling: CellNOpt, MaBoSS

Rodriguez-Mier *bioRxiv* 2024
Liu *NPJ Syst Biol Appl* 2019

Context specific networks

- Condition specific models
- Mechanistic insights
- Testable hypotheses

Software tools and environment

- Python with `uv` environment
- OmniPath database access: `omnipath` package
- Activity inference: `decoupler` package
- Causal reasoning: `corneto` package
- Data handling, visualization: `pandas`, `seaborn`

Available datasets

- Differential all omics (transcriptomics, phosphoproteomics, secretomics, proteomics; 7 time points)
- Activities: transcription factor & kinase
- Imaging data: collagen I deposition

Workflow overview - part 1

1. TF activities from transcriptomics
   - Univariate linear model (ulm) by `decoupler`
   - Prior-knowledge: TF regulons from CollecTRI
2. Data and PKN preparation
   - Select the times and modalities
   - Filter to significant changes
   - Causal network from database (OmniPath)
   - Filter the network to expressed genes
   - Prune network to reachable nodes
   - Prune data to network
3. Visualize activities and secretome

Workflow overview - part 2

1. Infer early network
   - `carnival` causal reasoning implementation by `corneto`
   - Prior-knowledge: human signaling network from OmniPath
   - Perturbations (inputs): early activities + TGF-beta
   - Measurements (outputs): early secretome
2. Infer late network
   - Perturbations (inputs): early secretome + TGF-beta
   - Measurements (outputs): late activities
3. Visualize and compare the context specific networks with activities color coded

Longitudinal multi-omics

TGFβ $\rightarrow$ early signaling $\rightarrow$ early secretome $\rightarrow$ late signaling $\rightarrow$ late secretome

Each arrow is a network inference

Project materials

https://github.com/saezlab/corneto-ebi-multiomics